

LEVEL 1: Master Unit Testing Registry

Series 100: User Authentication, Security & Session Management

Focuses on crypto hashes, formatting rules, token lifetimes, and gatekeeping rules.

This registry defines isolated unit tests for authentication and security components. Pages 2–10 document the **PHINMA Cagayan de Oro College (COC) OMR** system: Flutter mobile app, Laravel API, and Next.js web portal.

Unit ID	Target Function / Code	Method	Testing Scenario Profile	Expected Isolation Output
UT101	validateEmailFormat(email)	Input contains structural typos (e.g., test@com, @@domain.com).	Invalid email structure	Return false (Rejected)
UT102	validatePasswordStrength(pwd)	Input is less than 8 characters or misses a special character.	Weak password	Return false (Weak password)
UT103	hashUserPassword(plainText)	Raw string passed into cryptographic library (BCrypt).	Registration hash	Return unique 60+ char secure salt hash
UT104	verifyPasswordMatch(plain, hash)	Login password attempt against database hash.	Credential verify	Return true if match; false if modified
UT105	generateSessionToken(userId)	User authenticates successfully; token needs generation.	Session issue	Return structured JWT or secure random state string
UT106	isTokenExpired(timestamp)	Token expiry timestamp older than current server time.	Expiry check	Return true (Trigger auto-logout routine)
UT107	sanitizeHtmlInput(userInput)	Input contains unsafe script tags.	XSS attempt	Return stripped or encoded clean string

COC OMR implementation details for UT101–UT107 appear on Page 2.

COC OMR — Unit Testing (Series 100)

Stack: Laravel API (coc-omr-api) · Flutter app (omr_app) · Next.js web (omr_web)

Unit ID	Target Function / Code	Testing Scenario Profile	Expected Output (COC OMR)
UT101	Laravel email rule; Flutter _isValidEmail()	test@com, @@domain.com, blank email	Rejected; message: "Enter a valid email address."
UT102	PasswordRules (Dart/TS) + Laravel Password::defaults()	abc123, abcdefgh, Password1 (no symbol)	Rejected on register/reset; min 8 + letter + number + symbol
UT103	Laravel Hash::make() (bcrypt)	New teacher password at registration	60+ char bcrypt hash stored; plain text never saved
UT104	Laravel Auth::attempt()	Correct vs wrong password at login	true if match; false + "credentials do not match"
UT105	Sanctum createToken() — Bearer token (not JWT)	Successful login (web: device_name web; app: mobile)	Sanctum personal access token returned
UT106	Web: WEB_SESSION_MAX_AGE_SECONDS (7 days cookie). App: no timed expiry	Web session after 7 days; app token after 7 days	Web: re-login required. App: token valid until sign-out/revoke
UT107	PersonName.normalize() (PHP/Dart/TS) — replaces generic XSS test	MARIA DE LA CRUZ; Santos, Maria; extra spaces	Maria de la Cruz; Maria Santos; clean First Last format

Automated tests: test/password_rules_test.dart · test/person_name_test.dart · omr_web/test/person-name.test.ts · coc-omr-api/tests/Unit/PersonNameTest.php

LEVEL 2: Integration Testing Registry

Series 100: Frontend ↔ Backend API (COC OMR)

ID	Subsystems	Scenario	Expected Behavior
IT-101	Register UI ↔ RegisterController	Teacher submits name, @phinmaed.com email, COC department, password	POST /api/register; name normalized; pending admin approval; verification email
IT-102	Login UI ↔ Session storage ↔ LoginController	Approved teacher signs in	Token returned; web cookie coc_api_token; app SharedPreferences + offline PIN setup
IT-103	Results/Classes search ↔ Client filter	Partial name or OMR ID typed in search	Local grid filter without full reload (no debounced live API)
IT-104	Roster import ↔ ImportService/roster.ts ↔ StudentSyncController	CSV/Excel Class_List_Report uploaded	Parsed client-side; JSON sync to API (not multipart image upload)

LEVEL 3: System Testing — Series 100

Authentication, Security & Account Management

ID	Procedure	Test Data	Expected Response
ST-101	Register with email already in database	Existing @phinmaed.com	Pending: "Waiting for admin approval." Approved duplicate: "Use Login."
ST-102	Login with wrong password	Valid email, wrong pass	"Email or password is incorrect."
ST-103	Type in password field	Any input	Masked as •••••
ST-104	Register with blank required fields	Empty name/dept/password	Inline validation errors; submit blocked
ST-105	Teacher navigates to /dashboard/admin	Non-admin account	Redirected; access denied to admin desk
ST-106	Logout then browser Back	Active web session	Cookie cleared; dashboard requires re-login

LEVEL 3: System Testing — Series 200

Roster, Answer Keys & Student CRUD

ID	Procedure	Test Data	Expected Response
ST-201	Import roster CSV/Excel; confirm	Class_List_Report sample	Success counts; sections/students in Classes view
ST-202	Answer key: invalid numeric field	Passing score: abc	Validation error; save blocked
ST-203	Edit student name; Sync Now	Updated name	First Last format; visible on web after sync
ST-204	Remove student from roster	Existing row	Student removed after confirm/save
ST-205	Results with many scans; filter	50+ records	List loads; client-side filter (no server pagination UI)

LEVEL 3: System Testing — Series 300 & 400

Search, Filtering, Exports & Grading

Series 300 — Search & Filtering

ID	Procedure	Test Data	Expected Response
ST-301	Search Results/Classes by name or OMR ID	Known student	Only matching rows shown
ST-302	Search nonsense keyword	xyz987abc	Empty table; "No matching records"
ST-303	Filter by section and subject	Dropdown filters	Filtered scans and item analysis

Series 400 — Reporting & Computations

ID	Procedure	Test Data	Expected Response
ST-401	Scan practice sheets vs manual grading	10-sheet validation set	≥95% match; partial credit correct
ST-402	Export OMR PDF or results PDF	Print / Export click	Valid PDF at 100% scale
ST-403	Export results CSV	Export CSV	Opens correctly in Excel

LEVEL 3: System Testing — Series 500

Sync, Configuration & UI Feedback

ID	Procedure	Test Data	Expected Response
ST-501	Login while cloud API is waking (Render)	Slow first request	"Connecting to school server..."; retries; no false account error
ST-502	Open Settings; view account/sync	Logged-in teacher	Name, COC school, department; Sync Now works online
ST-503	Scan offline; reconnect; Sync Now	Airplane mode exam	Local scans preserved; cloud updated after sync

Template vs COC OMR (panel notes)

Generic template item	COC OMR actual
JWT tokens	Sanctum Bearer tokens
Live debounced API search	Client-side filter
Multipart image upload	CSV/Excel roster import
Table pagination	Limited — search/filter used
Mobile timed auto-logout	Web cookie 7 days only

LEVEL 4: User Acceptance Testing (UAT)

PART 1: Participant Profile Matrix — COC OMR

Participant ID	Role	Organization	Date	Signature
UAT-EX-01	IT / Capstone Adviser (Expert)	PHINMA COC — IT/CS	___/___/2026	
UAT-EX-02	COC OMR School Admin	PHINMA COC — Admin desk	___/___/2026	
UAT-EX-03	Department Head / Lead Teacher	COC — COE/SCCJ/CMA/CIT/CEA/CAHS	___/___/2026	
UAT-EU-01	Classroom Teacher (End-User)	COC — assigned department	___/___/2026	
UAT-EU-02	Classroom Teacher (End-User)	COC — different department	___/___/2026	
UAT-EU-03	Teacher — phone scan + web prep	COC	___/___/2026	

Minimum: 3 domain experts + 10–15 teachers completing prep → print → scan → sync.

PART 2: ISO/IEC 25010-Aligned Survey (COC OMR)

Likert: 5 Strongly Agree · 4 Agree · 3 Neutral · 2 Disagree · 1 Strongly Disagree

#	Criterion	5	4	3	2	1
1.1	Functional Completeness: register, admin approval, roster, keys, print, scan, sync, export					
1.2	Functional Correctness: scan scores match answer keys					
1.3	Functional Appropriateness: offline exam day + web prep workflow					
2.1	Login vs Register vs PIN understood without developer					
2.2	New teacher learns roster import + print after short demo					
2.3	Scanner, Review, Sync easy on typical Android phone					
2.4	Web UI professional (COC green/amber branding)					
2.5	Text and controls readable on phone and laptop					
3.1	Scanning responsive during class session					
3.2	Stable on mid-range phones; offline usable					
4.1	Non-admins cannot access Admin → Access control					
4.2	Teachers see only their own cloud data					
4.3	Admin approve/revoke tied to logged-in admin					
5.1	Low-confidence scans flagged for Review					
5.2	Offline PIN works; sync recovers after Wi-Fi					
6.1	Web on Chrome/Edge; app on target Android					
6.2	Deployable to school server without workflow change					

PART 3: Formal Client Acceptance & Sign-Off

This signature sheet is the definitive evidence required by capstone panels. It proves the capstone project solved a real institutional need and has been officially handed over.

Acceptance Statement:

"By signing below, I acknowledge that I have actively participated in or reviewed the User Acceptance Testing (UAT) phase for the developed system. I confirm that the functional deliverables, user interfaces, security protocols, and operational workflows meet our agreed specifications and are accepted for live deployment."

Approved By:

Client Printed Name & Designation

Signature over Printed Name

Date Signed: ____ / ____ / 2026